

RESEARCH ARTICLE



Land as a discretionary extra-budgetary revenue tool: land reserve system, monopoly supply power and local government finance

Xuewen Li^a, Liurulin Ji^a, Hongyang Li^a and Lianying Yao^{b,c}

^aSchool of Public Affairs, Zhejiang University, Hangzhou, Zhejiang, China; ^bInstitute for Regional and Country Studies, Zhejiang University of Technology, Hangzhou, Zhejiang, China; ^cSchool of Economics, Zhejiang University of Technology, Hangzhou, Zhejiang, China

ABSTRACT

This article conceptualizes the Land Reserve System (LRS) as an institutional fiscal instrument through which Chinese local governments transform land into discretionary extra-budgetary revenue under post tax-sharing fiscal pressures. Leveraging a panel of 218 prefecture-level cities (1999–2013), we assess whether and how LRS adoption restructures local fiscal revenue composition. The empirical strategy combines a Difference-in-Differences framework with an identification approach that exploits the exogenous fiscal shock induced by the 2002 corporate income tax reform and an instrumental variable based on Cox hazard-predicted adoption timing. We find that adopting the LRS leads to a large and persistent increase in land-finance dependence, measured by the share of land transfer revenue in the general public budget revenue. Mechanism analyses suggest that the LRS enables more land for commercial and service uses, consistent with strategic supply management, thereby increasing housing prices and reinforcing land based fiscal extraction.

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1. Introduction

Local government bureaucratic elites are always motivated to pursue discretionary budgets (Migué et al., 1974; Niskanen, 1975). Especially in East Asian countries during their economic takeoff, discretionary budgets were closely linked to the actions of local governments committed to promoting economic growth within their jurisdictions, ultimately leading to the formation of a developmental state (Öniş, 1991). In such contexts, local governments often resort to policy instruments that mobilize quasi-fiscal resources outside regular budgetary channels, since these tools can be deployed rapidly, targeted strategically, and aligned with performance-oriented administrative incentives. China serves as a prominent example of this broader political – fiscal logic. Following the 1994 tax-sharing reform, the retention of profits by enterprises as an extra-budgetary revenue source was cut off (Li et al., 2012). As a result, prefectural and municipal governments faced a persistent mismatch between their revenue capacity and spending obligations (Jia et al., 2014; Wang & Ma, 2014). At the same time, the cadre evaluation system continued to reward investments and visible development outcomes. Under these conditions, local governments increasingly turned to land transfer revenue as a new extra-budgetary revenue channel, giving rise to the widely discussed land finance model (Cao et al., 2008; Huang & Chan, 2018; Wu et al., 2015). Yet an analytically prior question remains under-specified in the public policy literature. What institutional arrangement makes land systematically obtainable as a discretionary, extra-budgetary revenue fiscal tool, and how does that arrangement reshape the structure of local public finance and the governance strategies built upon it?

A large body of research has examined the rise of China's land finance and its consequences for investment-led growth, governance distortions, and housing outcomes. This paper clarifies that the proceed from local government sales of state-owned land-use rights matters not only because it is sizable, but also because it can function as a comparatively discretionary fiscal resource under China's intergovernmental system (Cao et al., 2008; Gyourko et al., 2022; Huang & Chan, 2018). Fiscal and political incentives further reinforce this reliance. The cadre evaluation system's emphasis on economic performance encourages local

governments to prioritize land-based urban expansion and development projects as visible signals of achievement (Han & Kung, 2015), contributing to a local developmental governance pattern in which land, infrastructure, real estate, and industrial planning become tightly bundled (Su & Tao, 2017; Tao et al., 2010). At the same time, the governance costs of land-based fiscal strategies are becoming increasingly evident. Land market processes often fail to meet good governance principles, including transparency and participation, while administrative interventions in land supply can lead to misallocation and social tensions (Koroso et al., 2013; Wu & Heerink, 2016). Empirically, restrictive land-use controls have been associated with higher housing prices (Zhang et al., 2013), and land revenue windfalls may create incentives consistent with a political resource curse, including rent-seeking and corruption risks (Chen & Kung, 2016). These findings underscore land's centrality to local public finance and urban governance, but they do not fully resolve the institutional question of how land becomes reliably usable as an extra-budgetary revenue fiscal instrument.

This article argues that the Land Reserve System (LRS) is the key institutional technology that converts land from a passive asset into a systematically controllable fiscal tool. Without LRS, land conveyance can remain fragmented, project-driven, and administratively dispersed across agencies and quasi-governmental entities, limiting the state's capacity to coordinate acquisition, preparation, timing, and pricing in a coherent manner. LRS institutionalizes a chain of actions encompassing land acquisition and collection, consolidation, servicing and preparation, reservation, and unified release, thereby centralizing operational authority and organizational capacity for managing the primary land market (Tian & Ma, 2009; Wang et al., 2020). By strengthening the government's monopoly position in the primary land supply, LRS increases discretion over when and what land to release, enabling intertemporal supply management and policy-relevant price effects (Sasu et al., 2022; Wu, 2022). In this sense, the LRS is not just a technical arrangement for land administration but an institutional mechanism that enhances the state's capacity for value capture and fiscal leverage.

This perspective implies a claim central to the paper's contribution that land is not inherently an extra-budgetary revenue fiscal instrument and becomes one through LRS-enabled state capacity in the primary land market. LRS enhances local governments' ability to (i) centralize and standardize primary-market supply through a single administrative gateway, (ii) manage the timing and structure of supply to influence prices and revenue flows, and (iii) institutionalize value capture in ways that render land transfer revenue more predictable and scalable in fiscal practice (Tian & Ma, 2009; Wu, 2022). Importantly, the same institutional capacities that facilitate fiscal extraction can also reshape allocation outcomes. A growing literature emphasizes that public finance considerations can shape land supply strategies, including industrial land discounts and zoning or supply decisions that prioritize fiscal returns (He et al., 2022; Henderson et al., 2022). LRS, by strengthening centralized control over the supply pipeline, provides the organizational infrastructure through which such fiscally motivated land allocation strategies can be executed.

Existing scholarship has addressed aspects of this argument, yet two major gaps remain. First, credible causal evidence on how LRS adoption reshapes local fiscal revenue structures, particularly dependence on land-related income, remains limited. Endogeneity is a core concern as jurisdictions facing greater fiscal stress may be more likely to adopt LRS, and the same stress may independently increase land reliance (Han & Kung, 2015). Second, the mechanism linking LRS to downstream governance outcomes is not sufficiently tested. If LRS deepens land-based fiscal dependence, it likely does so not only by increasing transaction volumes, but also by reshaping the composition of land supply towards higher-premium uses. Prior studies document strategic land supply behaviour such as land hoarding, price manipulation, and shifts in land-use structure that can increase fiscal returns (Du & Peiser, 2014; Lichtenberg & Ding, 2009; Tao et al., 2010). Yet we still know too little about whether LRS, as an institutional reform, systematically shifts local governments towards a more high-end land-use structure, and whether such structural changes contribute to higher housing prices and, in turn, reinforce dependence on land-based fiscal revenue.

Accordingly, this paper addresses three interrelated research questions. Q1: Does the establishment of an LRS increase local governments' dependence on land-based revenues, as measured by the share of land transfer revenue in the general public budget revenue? Q2: Does LRS affect housing prices through a land supply structure channel, by reallocating land for commercial and service uses while reducing land for ordinary commercial houses and economically affordable houses? Q3: Is LRS adoption itself a fiscal response mechanism, triggered by exogenous revenue shocks that compress local fiscal space?

Empirically, the study constructs panel data for 218 prefecture-level cities from 1999 to 2013 and estimates a Difference-in-Differences (DID) framework comparing adopting and non-adopting cities over

time. To address adoption endogeneity, we exploit the exogenous fiscal shock induced by the 2002 corporate income tax reform, which altered the central-local sharing arrangement and generated heterogeneous revenue losses across cities depending on pre-reform exposure. Building on this shock, we construct an instrumental variable derived from a Cox proportional hazards model that predicts adoption timing under fiscal compression, thereby isolating the component of LRS adoption plausibly driven by externally induced fiscal losses rather than by unobserved land-market trajectories or local political preferences.

The paper contributes to the literature in three ways. First, it offers a public-finance-centred conceptualization of LRS as an institutional technology that makes land systematically obtainable as a discretionary extra-budgetary revenue instrument, complementing incentive-based accounts of why local governments seek land revenue (Cao et al., 2008; Huang & Chan, 2018; Wu et al., 2015). Second, it provides causal evidence on the fiscal consequences of LRS adoption, directly addressing a key identification gap in existing research (Wang et al., 2023; Wu, 2022). Third, it connects fiscal institutional change to distributionally salient urban outcomes by testing the land-supply-structure mechanism and its housing price implications, thereby linking intergovernmental fiscal politics to the normative agenda of inclusive urban development and good governance (Koroso et al., 2013; Zhang et al., 2013).

The remainder of the paper proceeds as follows. Section 2 is literature review. Section 3 is institutional background. Section 4 introduces the data and variables. Section 5 presents the empirical strategy, including DID and the instrumental variable approach based on the 2002 fiscal shock and hazard-model-predicted adoption. Section 6 reports the main results and mechanism evidence. Section 7 provides heterogeneity and robustness checks. Section 8 concludes with policy implications for fiscal sustainability, land governance, and housing affordability in rapidly urbanizing Asian contexts.

2. Literature review

2.1. Fiscal decentralization and the rise of extra-budgetary fiscal strategies in China

In the 1980s, the fiscal contracting system ('Fen Zao Chi Fan') expanded local fiscal autonomy and strengthened marginal incentives for local revenue retention in China. In practice, local governments often sought to maximize local interests through both revenue expansion and revenue retention strategies. Classic accounts show that localities could protect local enterprises and segment markets ('local protectionism') to expand the local tax base, while also retaining revenues that should have been remitted upward through various institutional channels (Oi, 1992; Wong, 1991; Young, 2000). The prevalence of regional market segmentation and local protectionism has been documented as a systemic byproduct of decentralization under incomplete market integration (Bai et al., 2004; Young, 2000).

Crucially, the fiscal contracting era also fostered extra-budgetary practices. Retained earnings within enterprises and locally controlled funds (often outside formal budget constraints) helped local governments finance development priorities with considerable discretion (Oi, 1992; Wong, 1997). Under the incentives of fiscal autonomy, local governments have made efforts to retain fiscal revenues that should have been transferred to the central government through methods such as concealment and artificially lowering fiscal growth rates. By doing so, they leave the tax sources within the taxable economic entities, enabling enterprises to evade or reduce taxes. These revenues are then transformed into extra-budgetary revenue that can be freely controlled by local governments through other channels, such as transferring profits and management fees from township enterprises, fundraising, allocation, and mandatory donations. This is referred to as enterprise profit retention (Sun & Zhu, 1993). Huang and Wang (1993), in their investigation of extra-budgetary revenue in a northern province, found that in 1992, about 70% of the province's extra-budgetary revenue funds came from profit retention by state-owned enterprises and funds. This led to a continuous decline in the share of central government fiscal revenue in total fiscal revenue and GDP. However, the 1994 tax-sharing reform re-centred fiscal authority and strengthened central revenue capacity, while local governments continued to bear substantial expenditure responsibilities. This institutional arrangement generated a more persistent form of vertical fiscal imbalance (Wong, 2000; Dollar & Hofman, 2006). The previous enterprise retention style of extra-budgetary resource mobilization became increasingly constrained under the evolving tax and fiscal administration regime. This institutional shift pushed local governments to seek new, institutionalized channels of discretionary revenue

that were more compatible with the post-1994 fiscal order and less directly subject to central revenue sharing (Sun & Zhou, 2014).

The literature has provided strong explanations for why local governments seek discretionary resources, highlighting factors such as fiscal stress, soft budget constraints, and growth-oriented incentives under decentralization (Montinola et al., 1995; Qian & Weingast, 1997; Wong, 2000). However, a key gap remains. Studies often stop short of fully exploring the incentive story and rarely treat specific policy instruments as fiscal response mechanisms that can be causally identified. In particular, existing literature seldom links fiscal shock, institutional adoption, and fiscal revenue structure into a unified, testable causal chain. This gap is central to the present study, which, rather than discussing land finance as a broad trend, examines whether the Land Reserve System as an institutional innovation systematically increases the share of land transfer revenue in local fiscal resources.

2.2. The land reserve system (LRS) as institutionalized land banking for extra-budgetary revenue

The LRS can be understood as a government-led land banking institution that centralizes the acquisition, consolidation, preparation, and release of land. Discussions of land banking, both internationally and within Asia, highlight its governance functions, such as coordinating land assembly and redevelopment, providing serviced land, and enhancing the state's capacity to manage supply and capture value. At the same time, these discussions raise concerns about discretion, transparency, and distribution (Sasu et al., 2022; Tian & Ma, 2009). In China, the LRS is closely tied to the governance of the primary land market, where local governments play a hybrid role as supplier, regulator, and often de facto market operator.

To clarify the LRS's institutional significance, it is helpful to contrast land conveyance practices before the consolidation of a unified reserve platform. The legal framework for granting land-use rights was established early on, such as with the 1990 Interim Regulations on the Assignment and Transfer of the Right to the Use of State-owned Land in Urban Areas (State Council Decree No. 55) and the Urban Real Estate Administration Law, which outlines bidding, auction, listing and agreement as methods for granting land-use rights (Standing Committee of the National People's Congress, 1994; State Council of the People's Republic of China, 1990). However, in the absence of a strong, centralized reserve institution, the operational chain of land supply was often project-based and fragmented. This commonly included: (i) land expropriation and conversion followed by direct conveyance tied to specific projects; (ii) the activation of state-owned stock land held by state-owned enterprises (SOEs), public institutions, or the military through administrative adjustments, such as converting allocated land to granted land via supplementary procedures; (iii) bundled land supply in development zones, where zone authorities or local platforms effectively performed quasi-reserve or quasi-primary development functions with varying rules and reporting practices.

The LRS reorganizes these previously dispersed links into a unified pipeline. It centralizes land supply decisions, facilitates net land conveyance through standardized preparation, and builds an inventory of land assets. Together, these features strengthen local governments' capacity to manage the timing of land supply and respond to market conditions (Tian & Ma, 2009; Wang et al., 2020). A key policy milestone often associated with strengthening state capacity over land assets is the State Council's notice on strengthening the management of state-owned land assets (Guofa [2001] No. 15), which has been interpreted in the literature as part of an institutional push towards more standardized, market-oriented and government-controlled land conveyance frameworks.

Despite the descriptive and institutional accounts, much of the LRS literature either focuses on institutional history and functions or on correlational outcomes, while providing limited causal evidence on the mechanism. This is why our study targets by treating LRS explicitly as a fiscal-response institutional tool.

2.3. Land finance, local developmentalism and strategic land supply

A large literature documents the rise of land-based revenues in urban China and their centrality to local development strategies. Research on land conveyance, land price dynamics, housing markets, and supply elasticity shows that primary land market operations can generate substantial fiscal proceeds and shape urban investment patterns (Cao et al., 2008; Du & Peiser, 2014; Tao et al., 2010). Political economy accounts further highlight how land transfer revenue can interact with local incentives and governance arrangements,

contributing to investment-led growth and the entrenchment of developmental local states (Oi, 1992; Su & Tao, 2017).

At the same time, the governance consequences of land finance are widely discussed. Land-based fiscal strategies can be associated with debt risks and fiscal procyclicality, distortions in land allocation, and social concerns such as housing affordability and distributive tensions (Chen & Kung, 2016; Koroso et al., 2013; Zhang et al., 2013). However, much of this work treats land finance either as (i) an outcome variable driven by structural incentives, such as decentralization and competition, or (ii) a macro background to urban development, rather than identifying how specific institutional devices expand the government's operational policy toolkit and reshape the composition of fiscal revenues. In particular, the distinction between revenue scale and revenue structure matters for public policy. A system can become fiscally dependent on land even if land revenue is volatile, as long as land's share in fiscal resources rises and becomes hard to substitute. This motivates the paper to conceptualize land finance dependence as a feature of fiscal structure, measured by the share or weight of land transfer revenue in total local fiscal resources, rather than simply the absolute level of land transfer revenue.

3. Institutional background and theoretical framework

As mentioned above, in the land finance logic chain, the LRS plays a key role as one of the core mechanisms. After the tax-sharing system reform, local governments had strong incentives to seek new sources of extra-budgetary revenue. Under the land system arrangement in 1998, 30% of the land transfer revenue from newly added construction land went to the central government, while 70% went to local governments. Revenue from the sale of stock construction land, however, mostly went to local governments. Therefore, local governments had significant incentives to maximize land transfer revenue through land management (land finance). Against this backdrop, the LRS emerged as a key institutional mechanism that operationalizes local governments' control over land supply. The system was established and rapidly diffused across cities from the late 1990s to the early 2000s, with dedicated land reserve institutions (often titled Land Reserve Centers) becoming the organizational carriers for land acquisition, consolidation, servicing, and subsequent transfer. By converting land supply decisions into an agency-based and process-managed workflow, the LRS helps institutionalize both the pace and composition of land supply in the primary market, thereby strengthening local governments' capacity to manage land as a fiscal and development instrument (Tian & Ma, 2009; Wang et al., 2020). Table 1 documents this process of institutional diffusion. The number and share of prefecture-level cities that established land reserve centers increased sharply between 1999 and 2013. This expansion provides a clear institutional basis for examining how the LRS may reshape local fiscal behaviour.

Complementing the discussion of institutional diffusion, Figure 1 plots event-time patterns in the local governments' land finance ratio, defined as the ratio of land transfer revenue to general public budget revenue, around the year in which the land reserve center was established. The figure is based on data from the China City Statistical Yearbook (2000 to 2023). Following the classification of the National Bureau of Statistics of China, 70 large and medium-sized cities are categorized into first-tier, second-tier, and third-tier groups.¹ The horizontal axis represents years relative to the establishment of the Land Reserve Center (with year 0 denoting the year of establishment), while the vertical axis measures the ratio of land transfer revenue to general public budget revenue, reflecting the extent of local governments' fiscal dependence on

Table 1. Development of land reserve centers.

Year	Prefecture-level Cities with Established Land Reserve Centers	
	Number	Proportion of All Prefecture-level Cities
1999	14	5.9%
2001	98	41.5%
2003	166	70.3%
2005	180	76.3%
2007	184	78.0%
2009	195	82.6%
2011	217	91.9%
2013	220	93.2%

Note: The data source is publicly available government data, compiled by the author.

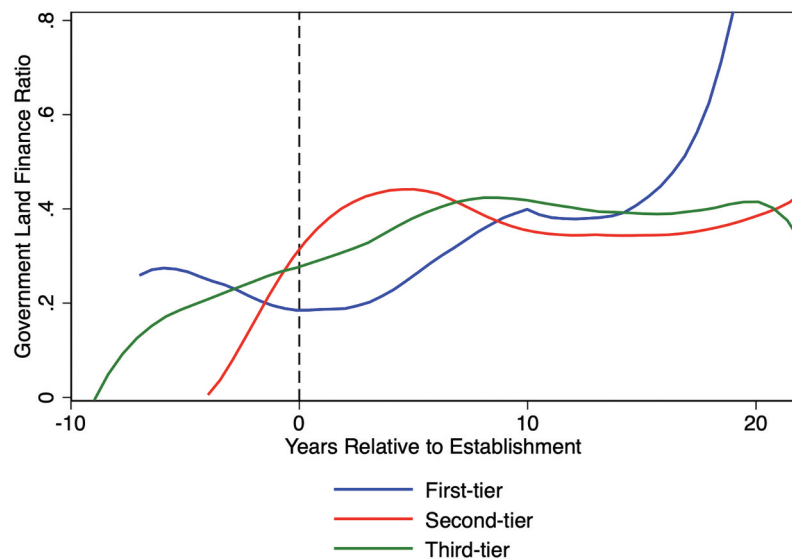


Figure 1. Dynamic change of local governments' land finance ratio before and after the establishment of the LRS.

land. Overall, fiscal dependence on land increased markedly across all three city tiers after the implementation of the LRS, indicating that the system strengthened local governments' centralized control over land supply and their fiscal incentives to monetize land assets. Among city tiers, first-tier cities exhibited relatively stable dependence before the policy but experienced a pronounced increase in the later years following its establishment. Second-tier cities recorded the sharpest and most immediate rise, showing a clear jump in government land finance ratio around the implementation period. Third-tier cities displayed a smoother but steadily upward trajectory over time. These patterns suggest that the LRS had heterogeneous effects across urban hierarchies. The differing performances among first-, second-, and third-tier cities are related to their respective capacities for local industrial development. Because first-tier cities are better able to attract net population inflows through industrial growth, they generate greater demand for land for commercial and service uses and land for residential use. Consequently, local governments in first-tier cities are more capable of using the land reserve system to control the structure of land supply and drive up housing prices, thereby obtaining higher land transfer revenue (Li et al., 2012). Specifically, it led to a long-term financialized expansion of land dependence in first-tier cities, a strong short-term fiscal stimulus in second-tier cities, and a gradual yet persistent increase in third-tier cities. Together, the results provide intuitive empirical evidence that the LRS reshaped local fiscal behaviour and contributed to the institutionalization of land-based finance in China.

Theoretically, The LRS reinforced local governments' monopoly over the primary land market, allowing them to strategically time land supply, manipulate prices, and adjust land conveyance methods (Sasu et al., 2022; Wu, 2022). More than a supply mechanism, the LRS serves as a vital fiscal tool for local governments. Figure 2 illustrates the market logic of controlled supply and demand creation adopted by local governments. The LRS enables local authorities to monopolize the primary urban land supply. To maximize fiscal revenues, governments transfer industrial land at low prices to attract manufacturing investment, thereby promoting industrial development and stimulating the agglomeration of population and capital (Li, 2025). This process creates effective land demand and facilitates short-term local economic growth (Tang et al., 2024). As industries and populations become more concentrated, market demand for land for commercial and service uses and land for residential use rises rapidly, while the supply of such land remains constrained. The resulting supply-demand imbalance drives housing prices upward. Ultimately, both land prices and land transfer revenue increase, further reinforcing local governments' dependence on land finance.

The 2016–2017 reform replacing land reserve loans with special-purpose bonds reduced hidden debt risks (Wang et al., 2023). The LRS also facilitates integrated governance by reallocating urban and rural construction land to advance urbanization (Tang et al., 2015). However, it has sparked debate over governance and equity, as centralizing land power may weaken market mechanisms, distort prices, and misallocate resources

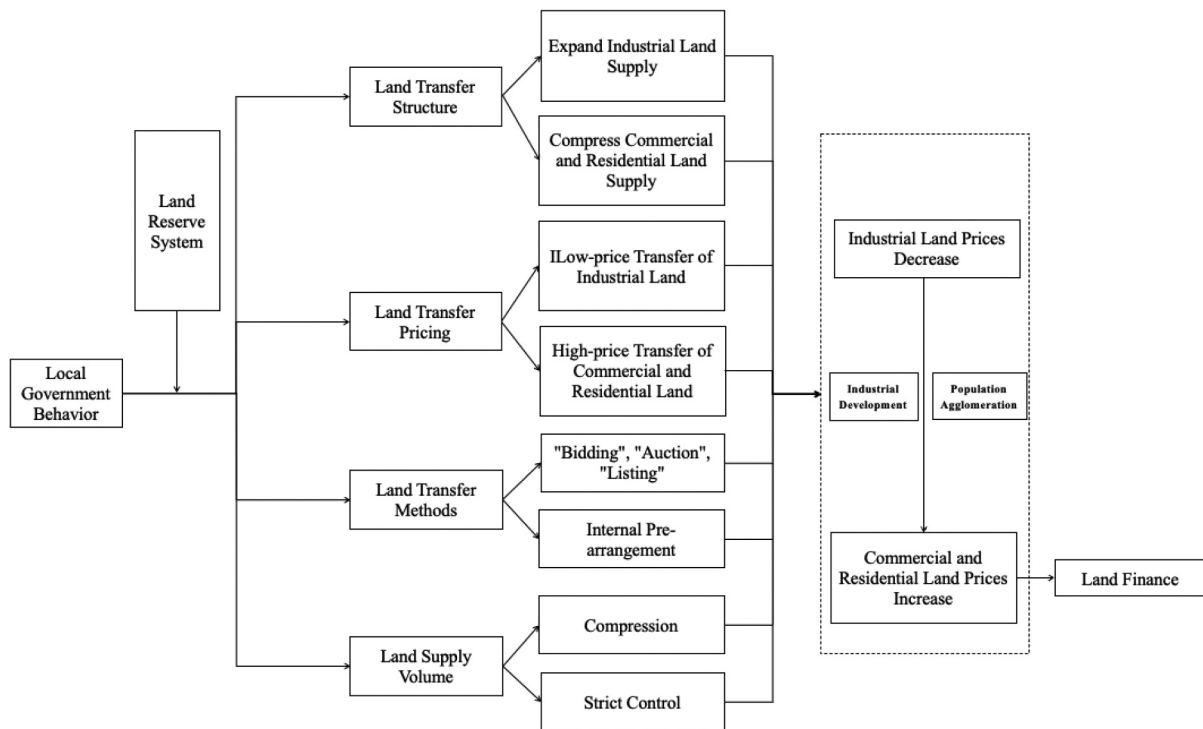


Figure 2. Operational mechanisms for local government behavior.

(Liu et al., 2016; Zhou et al., 2020). Thus, the LRS serves both as a regulatory tool and a lever for fiscal expansion and debt management. Another key function of the LRS is restructuring land supply. To maximize fiscal returns, local governments often prioritize land for commercial and service uses and land for high-grade residence while reducing low-priced industrial land.² (He et al., 2022; Henderson et al., 2022). Land transfer revenues on these parcels are closely linked to fiscal revenues, and financially pressured governments are more likely to release high-value plots (Lichtenberg & Ding, 2009; Tao et al., 2010). Such shifts affect not only fiscal income but also urban spatial patterns and industrial development (Cao et al., 2008; Du & Peiser, 2014). Converting industrial to commercial and service or residential land has become a key revenue tool (Han & Kung, 2015), and some governments hoard land as a fiscal asset, heightening market risks (Du & Peiser, 2014). Figure 3 provides a clear visual representation of the flow from land acquisition to final transfer, illustrating the steps that local governments take to manage land for fiscal revenue generation.

Overall, researches on land finance and the LRS have formed a mature framework explaining the interaction between land conveyance structures, supply dynamics, and government behaviour (Gyourko et al., 2022; He et al., 2022; Wu et al., 2015). Yet two gaps remain. First, a lack of rigorous causal identification on LRS impacts on fiscal dependence (Wang et al., 2023; Wu, 2022). Second, insufficient empirical analysis of how fiscal motives shape implementation strategies, such as limiting land for

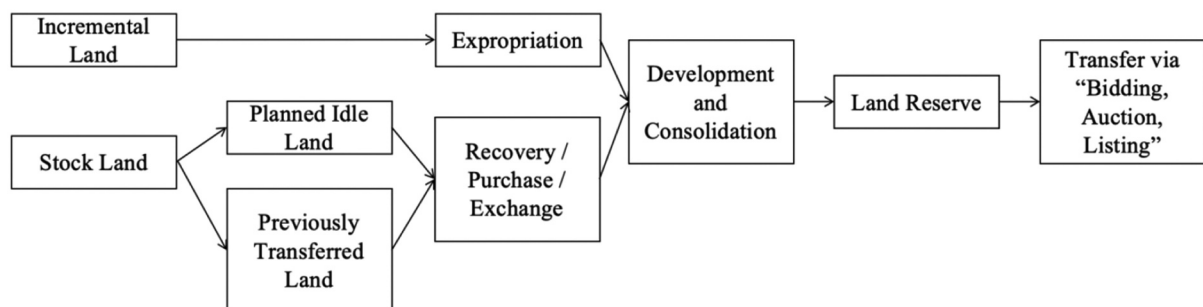


Figure 3. Operational mechanisms of the land reserve system.

ordinary commercial houses and economically affordable houses, prioritizing land for high-grade residence, and timing supply to achieve revenue goals (Chen & Kung, 2016; Du & Peiser, 2014). Therefore, this paper aims to empirically analyse the role of the LRS in enhancing local governments' reliance on land finance by constructing a DID model, using panel data from 218 prefecture-level cities between 1999 and 2013.

Across Asia, different institutional paths have emerged in land governance and urban development, reflecting each country's balance between state control and market coordination. In comparison with other Asian economies, Japan regulates its land market mainly through indirect instruments such as land readjustment, urban redevelopment, taxation, and land-use planning. Singapore, by contrast, follows a state-led integrated approach linking land management with housing and urban governance. The Housing Development Board system ensures affordable housing through multi-tiered supply and strong planning control (Luo, 2025). Strict housing standards further improve quality for low-income groups (Hanapi et al., 2023), while coordinated urban management achieves environmental efficiency (Zhou et al., 2022).

China's land system is unique. Urban land is state-owned, but local governments control the primary land supply, effectively creating a monopoly (Gyourko et al., 2022; Xu et al., 2009). Since 2008, local governments have expanded land financialization via land-backed loans and urban development bonds (Wu, 2022). The process of land and housing financialization in China is distinct from that in Western capitalist countries. By leveraging the state's monopoly over urban land use rights, China has driven the financialization of both the land and housing markets, which has contributed to rising land and housing prices (Chen & Wu, 2022; Y. Su et al., 2024). In this context, the financialization of land and housing has been significantly shaped by local governments' fiscal pressures. As local governments face increasing debt risks, they rely on land sales and land-backed loans to secure revenue, further entrenching the financialization of land (Lu et al., 2024). This grants local governments strong market power but also fosters fragmented, arbitrary governance (Lu & Wang, 2022). The regime reflects limited decentralization, with central authorities maintaining oversight while fiscally pressured localities use land management for revenue (Fan et al., 2020), and this bargaining dynamic permeates all aspects of land governance.

Building on this institutional foundation and the existing literature on China's policy evolution and empirical findings, this study conducts a theoretical analysis from the following perspectives.

From the perspective of fiscal federalism, since the 1994 tax-sharing reform, China's fiscal system has entered a stage of decentralized administration with centralized revenue distribution. While clarifying revenue-sharing rules and strengthening macroeconomic governance, the reform also tightened local budget constraints. At the same time, Local State Corporatism suggests that Chinese local governments act as entrepreneurial actors, mobilizing land and capital to pursue economic growth and political performance targets. With fixed administrative responsibilities but limited discretionary resources, local governments sought institutionalized alternative revenues, giving rise to a land-centred extra-budgetary revenue system land finance. Leveraging the higher demand for and appreciation potential of land for residential use and land for commercial and service uses relative to industrial land, often at the expense of industrial and housing land, improving short-term fiscal returns but worsening market imbalances and housing affordability. Within the framework of land financialization, the LRS represents a key mechanism linking fiscal incentives and land market behaviour. This system enables government monopoly over primary land supply, enhances timing and pricing discretion, and facilitates land value appreciation, thereby boosting related tax revenues. More than a technical tool of land management, it is embedded in China's fiscal governance logic, reflecting a shift from ad hoc responses to a strategic, sustainable extra-budgetary revenue regime.

Consistent with the causal chain, the framework yields three core hypotheses.

H1 (Fiscal-structure effect). Establishing an LRS increases local governments' dependence on land-based revenues, measured as the share of land transfer revenue in local fiscal resources, and this effect persists over time.

H2 (Operational discretion and land supply structure). LRS adoption shifts the composition of land supply towards higher-yield land uses, such as land for commercial and service uses, while reducing the relative supply of land for ordinary commercial and economically affordable houses. This pattern reflects

strengthened monopoly power and greater discretion over land supply in the primary market (He et al., 2022; Tian & Ma, 2009).

H3 (Price channel and feedback loop). Changes in land supply composition and greater discretion over supply timing raise land and housing prices, thereby increasing the fiscal proceeds extractable from land conveyance and further reinforcing dependence on land-based revenues, generating path dependence (Du & Peiser, 2014; Wu, 2022; Zhang et al., 2013).

4. Data and variable

4.1. Data sources

The data come from the China Urban Statistical Yearbook, China Land and Resources Yearbook, and China Urban Construction Yearbook, covering 218 prefecture-level cities from 1999–2013. Dates of land reserve center establishment were collected from local government websites. The 1999–2013 period is chosen for two reasons: (1) by 2013, 6.8% of cities lacked land reserve agencies, compared to only 2.1% in 2014, ensuring the DID model retains both treatment and control groups; and (2) national city land price data, from the Land and Resources Yearbook, are available from 1999 onward. This timeframe also captures the evolution of the LRS from its early stage to widespread adoption.

4.2. Dependent variables

The dependent variable in this study is the logarithmic form of *local government land finance dependence*. The original ratio variable is constructed as the ratio of city-level land transfer revenue to city-level general public budget revenue. The natural logarithm of this ratio is then taken. This indicator reflects the degree of local government dependence on land sales revenue and is an important quantitative measure of the strength of the land finance phenomenon. Compared to using absolute values of land sales revenue alone, local government land finance dependence better captures the relative position of land revenue in the local government's total revenue, effectively controlling for biases arising from differences in the fiscal size of different cities.

4.3. Independent variables

The key independent variable is *Lag of whether to establish LandReserveCenter*, indicating whether a city has implemented the LRS. The original variable equals 1 in the year of adoption and 0 otherwise, but given delayed policy effects, due to administrative adaptation and lagged land market responses, this study uses a lagged version. It equals 1 from the year the system begins to exert substantial fiscal and market impact onward, and 0 otherwise, aligning with the gradual effect of land policy and avoiding biases from anticipatory or short-term reactions. Since most adoptions occurred in the early 2000s, the variable remains stable after 2013, and is applied to the 1999–2013 period for estimation, mechanism analysis, and robustness checks.

To capture the economic and social impacts of the LRS, several variables are introduced. *Residential land intensity* measures the ratio of residential land area to built-up area, indicating the density of residential use. Proportion of land for ordinary commercial houses and economically affordable houses (*share_midlow*), with declines reflecting reduced affordable housing provision. Commercial-residential land ratio (*CommercialResidential_ratio*) is the ratio of land for commercial and service uses to total land for residential use, with higher values indicating a tilt towards high-value uses and stronger fiscal dependence.

4.4. Instrumental variables

To address the endogeneity issue of institutional adoption, this study uses survival analysis methods to construct an instrumental variable. The basic idea is to use the fiscal shocks that cities experienced prior to

adopting the LRS to estimate the likelihood and timing of their adoption. This approach allows the construction of an exogenous variable that is independent of the actual policy adoption process (Han & Kung, 2015). Following the 2002 corporate income tax reform, local governments experienced an abrupt and heterogeneous reduction in retained fiscal resources. In response, they could in principle adopt multiple strategies to restore fiscal capacity, such as enhancing tax collection, introducing new local fees, or lobbying for higher intergovernmental transfers. However, these alternatives were either institutionally constrained or produced delayed fiscal effects due to central control over tax authority and transfer allocation. By contrast, establishing a Land Reserve Center provided an immediate and administratively feasible channel for local governments to monetize land assets and collateralize future revenue streams. The approach treats the adoption of the LRS by a city as a failure event in the survival analysis. The Cox proportional hazards model is applied, with a simulated fiscal shock variable, *simshock*, derived from the corporate income tax reform as the main explanatory variable, controlling economic, population, and savings factors, to estimate the risk level of institutional adoption in each city. Based on the predicted values (risk scores) from the regression results, cities are divided into high-risk and low-risk groups, and a simulated adoption year variable, denoted as *adopt_sim_year*, is generated. A simulated adoption dummy variable, *set_sim*, is then defined, where *set_sim*=1 if the city's year is greater than or equal to the simulated adoption year, and 0 otherwise. This variable is used as an instrumental variable in the first-stage regression, simulating the time at which a city may adopt the system under fiscal pressure, thus helping to identify the exogenous impact of the LRS on land finance.

4.5. Control variables

To enhance explanatory power and mitigate omitted variable bias, this study includes several city-level controls in the baseline and mechanism regressions, capturing economic development, population structure, industrial composition, and financial resources. *Degree of openness* measures a city's capacity to attract external capital, calculated as the share of industrial output from non-local funded enterprises in GDP. *Urbanization rate* is the ratio of urban residents to the registered population, indicating population concentration and urban expansion. At the macro level, *GDP* and *Population density* represent economic scale and spatial population distribution, while *Average wage* reflects labour costs and income levels. *Industry output* value measures the manufacturing base and industrial land demand. *Resident wealth*, defined as the year-end savings balance of urban and rural residents (10,000 yuan) from the China Statistical Yearbook, indicates financial accessibility through household saving capacity, thereby affecting government financing capacity and land sales. *Fiscal gap* defined as the ratio of the difference between local budget expenditure and local budget revenue to local budget revenue, is used to capture the degree of local fiscal stress. All variables are log-transformed to reduce heteroscedasticity and enable coefficient interpretation, and are constructed from annual city statistical yearbooks to ensure consistency. Descriptive statistics are shown in the Table 2 below.

5. Model specification

5.1. Difference in differences specification

To enhance the validity of causal identification, this study employs a DID approach. By constructing event time windows before and after the implementation of the LRS, it examines whether the reliance on land based fiscal revenue differs significantly between the treatment group and the control group over time.

In order to systematically assess the impact of the LRS on local government dependence on land transfer revenue, we estimate the following baseline regression model:

$$Y_{i,t} = \alpha + \beta \cdot set_{i,t} + \gamma \cdot X_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t} \quad (1)$$

The variable $Y_{i,t}$ represents the degree of land finance dependence for city i in year t , and is specifically measured by the logarithm of the ratio of land transfer revenue to local general public budget revenue. The core explanatory variable $set_{i,t}$ is defined with a one-year lag in the actual adoption year, representing the

Table 2. Summary statistics.

Variable Name	Control Group (whether to establish LandReserveCenter = 0)					Treatment Group (whether to establish LandReserveCenter = 1)				
	mean	SD	min	max	Number	mean	SD	min	max	Number
<i>Dependent Variables</i>										
ln(Local government land finance dependence)	-2.59	1.46	-7.43	0.96	846	-0.84	0.92	-6.66	2.47	2392
ln(Per capita local government land finance dependence)	3.42	1.85	-2.77	8.71	846	6.21	1.56	-0.26	10.34	2392
<i>Independent Variables</i>										
Lag of whether to establish LandReserveCenter	0.00	0.00	0.00	0.00	852	1.00	0.00	1.00	1.00	2403
ln(House price)	7.33	0.46	5.12	8.93	394	7.85	0.57	6.14	10.10	2306
ResidentialLand_intensity	0.33	0.08	0.16	0.57	405	0.30	0.07	0.04	0.74	2117
<i>Instrumental Variables</i>										
simshock_avg	-0.09	0.06	-0.34	-0.01	849	-0.10	0.06	-0.34	-0.01	2376
dln_revenue	0.04	0.09	-0.29	0.23	849	0.04	0.09	-0.29	0.23	2376
<i>Control Variables</i>										
Degree of openness	0.12	0.22	0.00	1.50	798	0.23	0.31	0.00	1.93	2379
Urbanization rate	-1.50	0.90	-3.61	1.52	300	-1.65	0.79	-3.93	1.46	2276
lnGDP	14.78	0.87	12.04	17.72	852	15.77	0.99	12.67	18.85	2403
ln(Population density)	7.25	1.22	3.30	9.55	300	7.76	0.94	4.23	9.91	2276
Average wage level	9.18	0.45	8.31	11.35	852	9.99	0.56	2.71	11.83	2403
Industry output value	14.44	1.16	11.16	18.38	852	15.88	1.32	11.26	19.53	2403
ln(Resident wealth)	14.43	0.87	10.82	17.30	634	15.32	0.99	12.50	18.64	2403
Fiscal gap	0.99	0.93	-0.69	5.80	1059	1.24	1.04	-0.35	5.90	2185

Note: This table categorizes and summarizes data based on whether a land reserve center has been established (whether to establish LandReserveCenter = 0 or 1).

point in time when the policy begins to exert influence on fiscal outcomes. Control variables $X_{i, tr}$, μ_i represents city fixed effects, δ_t represents time fixed effects.

5.2. Instrumental variable approach

Since the adoption of the LRS may not be entirely exogenous, there is a risk of endogeneity bias. For example, local fiscal pressure may simultaneously drive both the adoption of the system and the reliance on land-based revenues, thus confounding causal inference (Han & Kung, 2015).

As shown in Figure 4, the 2002 corporate income tax reform substantially reduced the fiscal resources retained by local authorities, marking a key step in the recentralization of fiscal power. Due to the heterogeneous reliance on corporate income tax across cities before the reform, the actual intensity of the resulting fiscal shock varied considerably.

Following Han and Kung (2015), we construct a simulated instrumental variable (SIV) based on the pre-reform enterprise-tax exposure. Our innovation is to use the SIV to simulate the policy adoption timing and then derive a binary treatment indicator (*set_sim*) from that timing. Building on this institutional shift, we

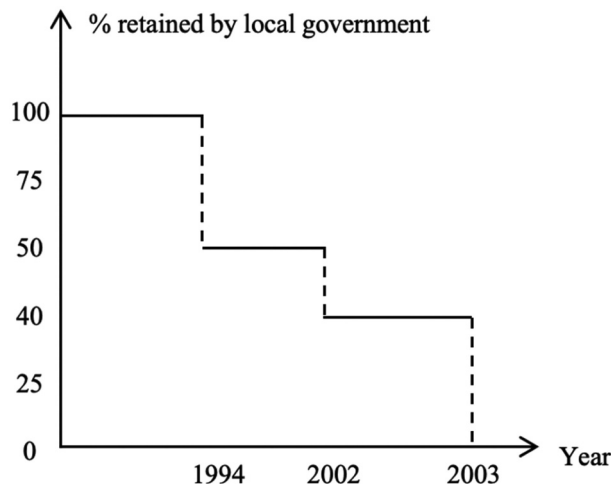


Figure 4. Changes in local government retention rate of corporate income tax, 1994–2003.

construct a simulated fiscal shock variable (*simshock_avg*) using the pre-reform share of corporate income tax in each city's local revenue. This variable captures the relative degree of fiscal compression resulting from the 2002 reform. Specifically, the simulated shock is computed as:

$$SIV_i = \frac{1}{2} \sum_{t \in \{2002, 2003\}} [\ln(I_{1,2001} + (1 - s_t)P_{i,2001}) - \ln(I_{i,2001} + P_{i,2001})] \quad (2)$$

Specifically, using 2001 as the baseline year, we define $I_{i,2001}$ as non-enterprise income tax revenue and $P_{i,2001}$ as corporate income tax revenue. We then compute the counterfactual change in total local revenue that would occur if the central government's statutory share of corporate income tax in 2002 ($s_{2002} = 0.5$) and 2003 ($s_{2003} = 0.6$) were applied to the pre-reform revenue composition. The SIV is calculated as the average of the simulated log-differences for these two years, thereby capturing the relative magnitude of fiscal compression each locality would have experienced under the reform.

We then estimate the timing of policy adoption using a Cox Proportional Hazards Model, treating system adoption as a failure event. The simulated fiscal shock (*simshock_avg*) serves as the key explanatory variable, while city-level economic and demographic characteristics are included as controls. The survival model is specified as follows:

$$h_i(t) = h_0(t) \cdot \exp(\beta_1 \cdot \text{simshock_avg}_i + X_i\beta) \quad (3)$$

Here, $h_i(t)$ denotes the institutional adoption hazard rate for city i at time t ; $h_0(t)$ represents the baseline hazard function; X_i is the vector of control variables.

Based on the risk scores predicted by the Cox model, cities are classified into high and low-shock groups. We then simulate the year of policy adoption (*adopt_sim_year*) for each city according to the estimated hazard distribution.

Using this simulated adoption timing, we construct a corresponding policy indicator variable, *set_sim_{i,t}*, which serves as the instrumental variable for actual adoption of the LRS.

$$\text{set_sim}_{i,t} = \begin{cases} 1, & \text{if } t \geq \text{adopt_sim_year}_i \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

5.2.1. First-stage regression model

In the first stage of the two-stage least squares (2SLS) estimation, we use the simulated policy indicator *set_sim_{i,t}* as an instrument to predict the actual adoption status of the LRS. The first-stage regression model is specified as follows:

$$Y_{i,t} = \alpha + \beta \cdot \text{set_sim}_{i,t} + X_{i,t}\vartheta + \mu_i + \delta_t + \varepsilon_{i,t} \quad (5)$$

Here, $Y_{i,t}$ denotes a one-year lag in the actual land reserve center adoption year, $X_{i,t}$ denotes the matrix of control variables, while μ_i and δ_t represent the city fixed effects and time fixed effects, respectively. $\varepsilon_{i,t}$ is the error term. This stage yields the fitted values of Lag of whether to establish LandReserveCenter, denoted $\hat{\text{set}}_{i,t}$.

5.2.2. Second-stage regression model

In the second stage, the predicted values obtained from the first stage $\hat{\text{set}}_{i,t}$ are used to replace the original endogenous variable. A regression is then conducted on the degree of local government land finance dependence.

$$Y_{i,t} = \gamma_0 + \gamma_1 \cdot \hat{\text{set}}_{i,t} + X_{i,t}\theta + \mu_i + \delta_t + \varepsilon_{i,t} \quad (6)$$

The dependent variable, $Y_{i,t}$, denotes the logarithmic ratio of land conveyance revenue to total local fiscal revenue for city i in year t , and is used to measure the degree of fiscal dependence on land. The simulated policy variable *set_sim_{i,t}* affects fiscal dependence only indirectly through its influence on actual policy adoption (*Lag of whether to establish LandReserveCenter*), but has no direct effect on the dependent variable.

Table 3. Baseline regression results.

Dependent variable: <i>ln (Local government land finance dependence)</i>	(1) Model 1	(2) Model 2
Lag of whether to establish LandReserveCenter	0.0987* (0.0519)	0.0971** (0.0482)
Controls	NO	YES
City FE	YES	YES
Year FE	YES	YES
Constant	−1.3420*** (0.0371)	2.9026 (1.8104)
Observations	3157	2481

Note: The control variables include Degree of Openness, Urbanization Rate, GDP, Population Density, Average Wage, Industrial Output, Resident Wealth and Fiscal Gap. Both specifications include city fixed effects and year fixed effects. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6. Empirical analysis

6.1. Baseline regression

To assess the impact of the LRS on local reliance on land-based revenue, we estimate baseline regressions (Table 3) using the log ratio of land conveyance revenue to general public budget revenue as the dependent variable and a lagged policy adoption indicator as the key explanatory variable. Column (1) excludes controls. Column (2) adds economic and social covariates. All models include city and year fixed effects with city-clustered robust standard errors. The policy coefficient is 0.0987 and significant at 10% without controls, and remains positive, rising to 5% significance with controls, indicating that the system significantly increases land fiscal dependence. These results provide initial evidence that the LRS is an important tool for easing fiscal pressure and boosting land-based revenue.

6.2. Mechanism identification

In this section, we examine how the establishment of the LRS increases local governments' fiscal dependence through its impact on land supply structure and housing prices. The LRS transforms local governments into the sole suppliers of urban land. With full monopoly power over primary land supply, local governments face strong fiscal incentives to maximize land transfer revenue. To achieve this, they strategically adjust the land supply structure according to market characteristics.

Specifically, different land-use types face distinct demand elasticities. Land for commercial and service uses and land for residential use are typically seller's markets with relatively inelastic demand, allowing governments to monopolize supply and set high land prices. By contrast, land for industry, mining and warehousing functions as a buyer's market, since capital mobility and inter-city competition force governments to lower land for industry, mining and warehousing prices to attract investment, promote industrial agglomeration, and stimulate employment growth. This industrial expansion, in turn, drives population agglomeration, which further elevates housing demand and pushes up residential prices.

Therefore, the LRS forms a two-step mechanism. In the first stage, by compressing the share of land for residential use and prioritizing the transfer of land for commercial and service uses with greater tax revenue potential, local governments create a structural bias in land supply, resulting in a decline in the proportion of land for residential use within total land supply. In the second stage, the reduction in land for residential use generates artificial scarcity in the housing market, directly pushing up housing prices. Meanwhile, the large-scale expansion of land for industry, mining and warehousing and land for commercial and service uses stimulates industrial clustering and population agglomeration, which further increases housing demand and amplifies the upward pressure on prices.

The regression results in Table 4 support this mechanism. The establishment of Land Reserve Centers significantly reduces the share of land for residential use supply. Meanwhile, land for residential use intensity is negatively correlated with housing prices, indicating that a decline in residential land supply leads to sustained housing price increases. In short, the LRS, by institutionalizing local governments' monopoly over land transfers, generates a supply-side mechanism that links land control, housing price escalation, and rising land fiscal dependence, thereby deepening the structural imbalance in urban land markets.

Table 4. Results of the analysis of land for residential use supply constraint channel.

VARIABLES	(1) Step 1: set ResidentialLand_intensity	(2) Step 2: ResidentialLand_intensity InHousePrice
Lag of whether to establish LandReserveCenter	−0.0095* (0.0050)	
ResidentialLand_intensity		−0.1782** (0.0800)
Controls	YES	YES
City FE	YES	YES
Year FE	YES	YES
Observations	2474	2457

Note: All regressions include control variables, city fixed effects, and year fixed effects. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

**Figure 5.** Changes in the ratio of land for commercial and service uses to land for ordinary commercial and economically affordable houses transferred by local governments, 2009–2013.

Based on Figure 5, we observe that from 2009 to 2013, all three tiers of cities experienced a clear upward trend in the ratio of land for commercial and service uses to land for ordinary commercial and economically affordable houses. The increase is modest in first-tier cities before 2011 but becomes pronounced afterwards, while second- and third-tier cities show sharper and earlier rises. These patterns indicate that following the implementation of the LRS, local governments in different tier cities increasingly prioritized land for commercial and service uses, reducing the supply of land for ordinary commercial and economically affordable houses and reflecting a systematic shift in land supply structure driven by fiscal incentives.

Overall, the trends of these three lines reflect the shift in land supply structure across different tier cities following the implementation of the LRS. Under fiscal pressure, local governments tend to allocate a larger share of land for commercial and service uses because such land yields higher land transfer fees and generates more tax revenue compared to land for residential use. By compressing the supply of land for ordinary commercial and economically affordable houses, the LRS has driven the expansion of land for commercial and service uses supply, which in turn has caused a shortage of land for residential use in the market and further pushed up housing prices.

6.3. Heterogeneity analysis

To further examine whether the adoption of the LRS varies with the degree of fiscal pressure, we conduct a subgroup survival analysis based on the Cox Proportional Hazards Model, investigating the effect of fiscal shock intensity on the timing of policy adoption. Figure 6 presents the predicted survival curves derived from the Cox model. These curves illustrate the evolution of the probability that a city has not yet adopted the LRS over time, following the corporate income tax reform. The horizontal axis denotes analysis time, beginning in 1999, while the vertical axis represents the survival probability, which refers to the likelihood that a city

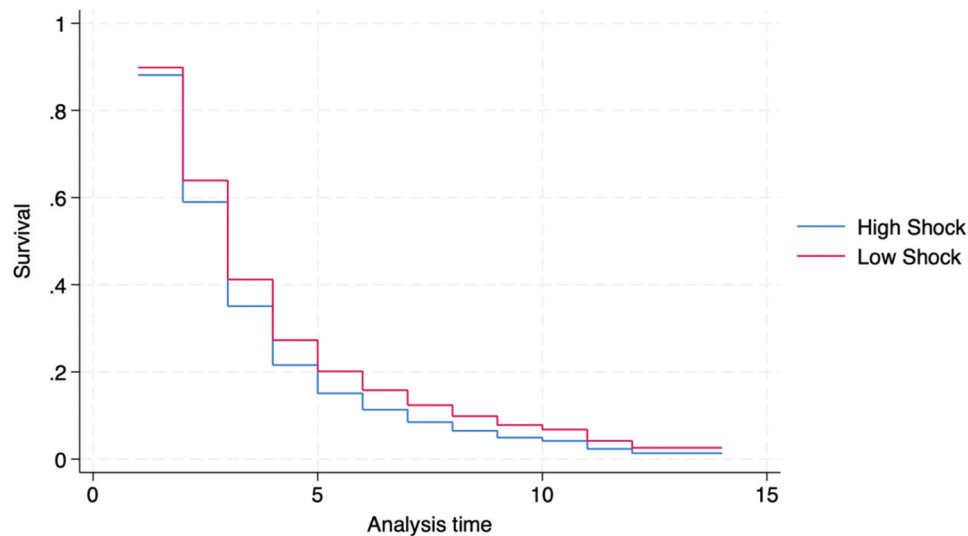


Figure 6. Predicted survival probability by shock intensity over time.

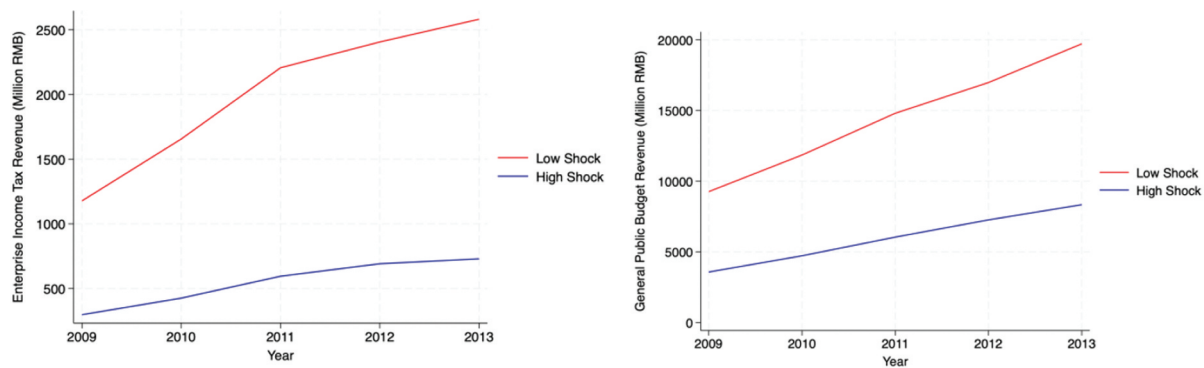


Figure 7. Trends in fiscal revenues by shock intensity.

remains outside the LRS by a given year. The survival curves clearly diverge, with cities in the high-shock group showing faster declines in survival probability, suggesting an earlier likelihood of LRS adoption. In contrast, cities in the low-shock group maintain the pre-reform institutional status for a longer period, suggesting weaker reform urgency and lower fiscal pressure. Figure 7 also reports the trends in fiscal revenues by different shock intensity.

These findings indicate that the simulated fiscal shock has a significant influence on the timing of institutional adoption, consistent with the theoretical expectation that fiscal incentives drive local policy reform. The time-differentiated responses captured by the survival curves further support the validity of *simshock_avg* as a plausibly exogenous instrumental variable for LRS adoption.

7. Robustness and extended analyses

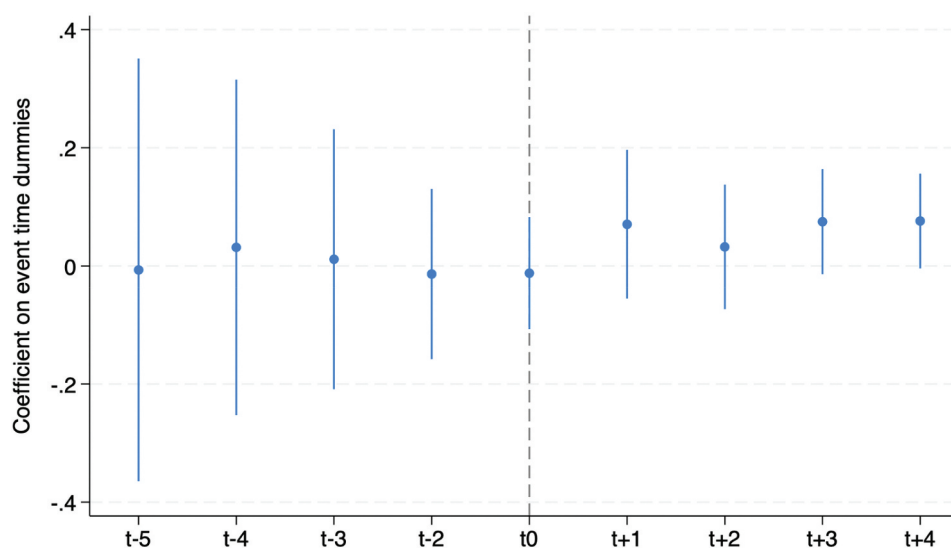
7.1. Parallel trends test and event study approach

To verify the identification assumption of the DID estimation strategy, we conduct an interaction trend test. The results are reported in Table 5. Specifically, we construct an interaction term between a post-policy time trend and the treatment indicator, defined as the product of the number of years since policy implementation and the treatment group dummy. This variable captures whether, prior to the implementation of the LRS, there were any differential trends in the outcome variable, namely local governments' dependence on land transfer revenue, between the treatment and control groups. The regression results indicate that the coefficient on the interaction term (*interaction*) is

Table 5. Parallel trends test.

VARIABLES	(1)
	DID Model with Interaction Term
interaction	−0.0157 (0.0126)
Constant	−1.198*** (0.0533)
Observations	3,178
R-squared	0.743

Note: This table reports the results of the parallel trend test using a DID model with an interaction term. The coefficient on the interaction term is statistically insignificant, suggesting that the treatment and control groups followed a parallel trend prior to the policy intervention. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

**Figure 8.** Event study plot of dynamic treatment effects.

statistically insignificant. This suggests that, in the pre-treatment period, there is no systematic difference in the outcome trends between cities that eventually adopted the LRS and those that did not. The parallel trends assumption is thus satisfied, providing a valid basis for the subsequent DID analysis. The regression includes both city fixed effects and year fixed effects to account for unobserved heterogeneity. Moreover, Figure 8, which presents the dynamic estimates from the event study analysis, visually reinforces the same conclusion by showing that pre-treatment effects fluctuate closely around zero without significant divergence.

7.2. Simulated IV and survival-based event study

To further assess the fiscal effects of the LRS and capture the dynamics of its causal impact, we employ an event study model using the simulated policy adoption year (*set_sim*). In Figure 9, the horizontal axis denotes event time, with $t = 0$ marking simulated adoption; the pre-adoption window ($t-5$ to $t-1$) shows estimated coefficients fluctuating around zero and confidence intervals largely crossing the zero line, indicating no significant pre-trend differences and supporting the parallel trends assumption. From $t + 1$ onward, coefficients display a steady and sustained upward trajectory, suggesting that the system consistently increases local governments' dependence on land-based revenue over time. This dynamic pattern not only confirms the robustness of the baseline estimates but also provides stronger causal evidence linking the institutional reform to long-term changes in local fiscal behaviour.

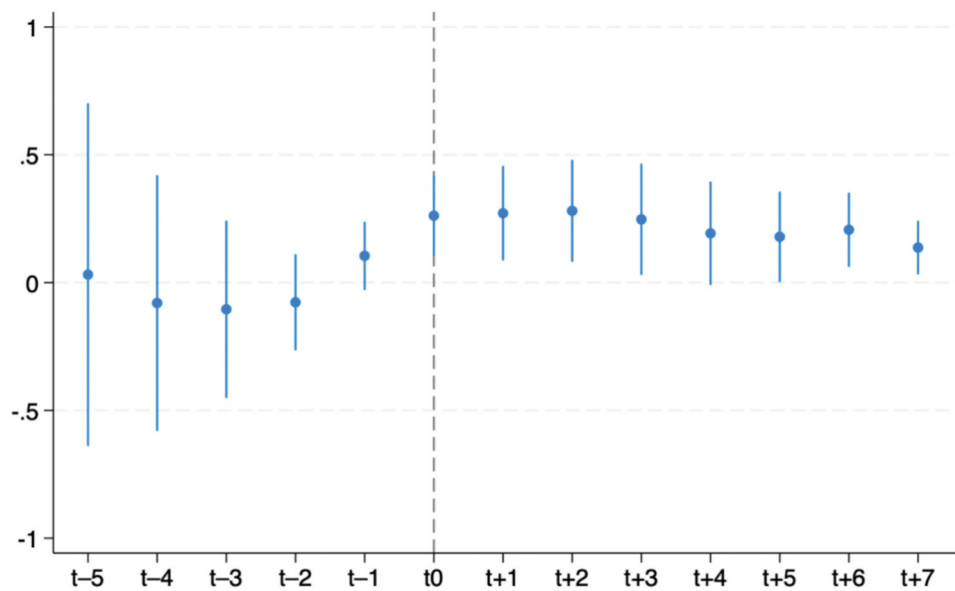


Figure 9. Dynamic effects of simulated policy adoption on land fiscal dependence.

8. Conclusion and discussion

This study systematically examines how the LRS influences Chinese local governments' fiscal dependence on land. Using panel data from prefecture-level cities, we employ fixed-effects regressions, mechanism analysis, and an instrumental variable strategy to identify both the existence and transmission paths of institutional effects. Results show that LRS implementation significantly increases reliance on land transfer revenue, even after controlling for population, industrial structure, and fiscal conditions. The LRS strengthens local governments' control over land allocation, enabling greater extraction of fiscal proceeds via land-based financing. The mechanism analysis reveals two profound impacts of the LRS. First, the system encourages local governments to reduce the supply of land for residential use, resulting in housing shortages and continuously rising regional housing prices. Second, it leads local governments to increase the proportion of land for commercial and service uses, thereby altering the urban land-use structure and shifting land allocation towards higher-yield uses. To address endogeneity, we use a simulated fiscal shock from the 2002 corporate income tax reform and construct an adoption-timing instrument via survival analysis. IV and event study results confirm significant and persistent fiscal effects. It is important to further clarify that the LRS does not directly determine the quantity of higher-level approved quotas for newly added construction land. Rather, it enhances local governments' capacity to manage effective primary-market land supply through centralized land collection, preparation, and unified release, particularly with respect to supply timing, composition, and spatial distribution, including stock-land redevelopment. Even in periods of relatively weak demand, local governments may still rely on selective land release and adjustments in land-use structure to stabilize land transfer revenue.

These results suggest that future reforms should better align fiscal incentives with sustainable land governance to address structural imbalances in urban development. First, to counter the LRS-induced preference for high-end development, housing supply quotas should be institutionalized within land reserve operations. Land reserve centers could be required to allocate a minimum share of land for residential use. Second, to curb the excessive expansion of land for commercial and service uses, urban land-use plans should impose upper limits on its proportions and maintain balanced shares for industrial and public-use land. Pilot programmes could encourage the conversion of underutilized land for commercial and service uses into land for industry, mining and warehousing or land for ordinary commercial houses and economically affordable houses. Third, since the LRS amplifies local fiscal reliance on land proceeds, performance- and resource-based intergovernmental transfers should be strengthened to compensate cities with limited land resources or high housing affordability pressures, thereby reducing the incentive for land-driven fiscal dependence. Collectively, these policy measures would realign local fiscal behaviour with sustainable land

governance objectives, curb speculative development incentives, and promote a more balanced and equitable urban growth model.

Compared with prior research that primarily relied on descriptive analysis without addressing endogeneity, this study advances the literature by introducing an instrumental variable approach and identifying the causal transmission channels of the LRS. Nevertheless, several limitations remain. First, while the descriptive statistics draw on city-level data from 1999–2023, the baseline and mechanism analyses are constrained by data availability and thus rely on 1999–2013 data, which may not fully capture recent changes in land financialization and fiscal reform. Second, although the simulated instrumental variable improves identification validity, concurrent policy or political factors may still introduce residual endogeneity. Third, the mechanism analysis mainly focuses on land-use restructuring and does not incorporate the role of land credit interactions that influence fiscal behaviour. Future research could employ micro-level or transaction-based data to explore institutional heterogeneity, offering deeper insights into how fiscal incentives reshape the governance logic of land and the trajectory of urban development in China.

Notes

1. The classification of cities into different tiers is based on the real estate price statistics for 70 major and medium-sized cities from the National Bureau of Statistics of China. First-tier cities include Beijing, Shanghai, Guangzhou, Shenzhen. Second-tier cities include 31 cities such as Tianjin, Shijiazhuang, Taiyuan, Hohhot, Shenyang, Dalian, Changchun, Harbin, Nanjing, Hangzhou, Ningbo, Hefei, Fuzhou, Xiamen, Nanchang, Jinan, Qingdao, Zhengzhou, Wuhan, Changsha, Nanning, Haikou, Chongqing, Chengdu, Guiyang, Kunming, Xi'an, Lanzhou, Xining, Yinchuan, and Urumqi. Third-tier cities include 35 cities such as Tangshan, Qinhuangdao, Baotou, Dandong, Jinzhou, Jilin, Mudanjiang, Wuxi, Xuzhou, Yangzhou, Wenzhou, Jinhua, Bengbu, Anqing, Quanzhou, Jiujiang, Ganzhou, Yantai, Jinan, Luoyang, Pingdingshan, Yichang, Xiangyang, Yueyang, Changde, Shaoguan, Zhanjiang, Huizhou, Guilin, Beihai, Sanya, Luzhou, Nanchong, Zunyi, and Dali.
https://www.stats.gov.cn/sj/sjkd/202509/t20250915_1961174.html
2. The Ministry of Land and Resources of the People's Republic of China classifies land use types into land for commercial and service uses, land for industry, mining and warehousing, and land for residential use. Residential land is further divided into land for high-grade residence, land for ordinary commercial houses, land for economically affordable houses, and cheap rent houses. Different land use types affect land prices, development models, and market structures, and also determine how local governments generate fiscal revenue through land sales and transfers.
According to the Current Land Use Status Classification (GB/T 21,010–2007), the definitions are as follows. Land for high-grade residence refers to construction-used land designated for high-grade residential construction during the reporting period, including land for residences with a floor area ratio (FAR) <1.0 and a building area of the residence > 144 m², as well as villas and high-grade apartments. Land for ordinary commercial houses refers to construction-used land designated for ordinary commercial house construction during the reporting period. Land for economically affordable houses refers to construction-used land designated for economically affordable housing construction. It includes land used for building houses through personal fundraising. Cheap rent house refers to construction-used land designated for cheap rent house construction.

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Notes on contributors

Xuwen Li, Associate Professor in the Department of Urban Development & Management at the School of Public Affairs, Zhejiang University, with research interests in the local government behavior, urban development and economic growth, as well as urban-rural integration and the spatial structural evolution of economic activities.

Liurulin Ji, a PhD student in the Department of Urban Development & Management at the School of Public Affairs, Zhejiang University, with research interests in the urban economics, urban-rural integration.

Hongyang Li, a PhD student in the Department of Urban Development & Management at the School of Public Affairs, Zhejiang University, with research interests in the computational social science.

Lianying Yao, Associate professor, ph. D., Institute for Regional and Country Studies & School of Economics, Zhejiang University of Technology. His research interests include digital governance and digital Economy.

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